

Interpretable Federated Kolmogorov–Arnold Surrogates for Autonomous and Evolutive Resource Optimization over Non-Stationary LEO Links

Phuc Hao Do¹ and Truong Duy Dinh²

Abstract—Low Earth orbit (LEO) networks must repeatedly solve resource-allocation problems whose optima have no closed form and that drift as satellites pass and interference regimes switch. We study an autonomous and evolutive approach in which a federated Kolmogorov–Arnold network (KAN) learns the *solution map* of an energy-efficiency transmit-power optimization, mapping a link state to its optimal power online. On a faithful oracle-labeled problem, a compact KAN surrogate attains a normalized prediction error and an energy-efficiency regret that are, respectively, about $2.3\times$ and $1.7\times$ lower than a parameter-matched multilayer perceptron (MLP) at equal parameter budget (Wilcoxon $p < 0.01$), while a linear predictor fails entirely, confirming the map is strongly nonlinear. Crucially, the additive spline structure of the KAN lets us *read off closed-form design rules* from the trained model, a transparency an MLP cannot provide. A contextual-bandit controller sparsifies the federated updates, halving the uplink volume while retaining the accuracy advantage over the MLP, tracing a favorable communication–accuracy frontier. We further report a transparent negative result: growing the spline grid online (self-evolution) does not improve over a well-sized static grid in this federated streaming regime. The study positions interpretable KAN surrogates as a practical building block for autonomous optimization in non-stationary networked-AI systems.

Index Terms—Kolmogorov–Arnold networks, federated learning, learning to optimize, LEO satellite networks, energy efficiency, interpretability, non-stationary environments.

I. INTRODUCTION

NEXT-generation non-terrestrial networks built on LEO constellations operate under fast, non-stationary conditions: the channel gain sweeps with elevation across a pass, and handovers between satellites and beams switch the interference regime on a timescale of seconds [1], [2]. Each terminal must continually re-solve resource-allocation problems such as energy-efficient power control, whose optimizers have no closed form and must be recomputed as conditions drift. Solving these from scratch online is costly, and the time-varying feasible region defeats precomputed lookups [3].

A learning-to-optimize view replaces the per-instance solver with a surrogate that maps a problem state directly to its optimal decision [3]. For this to be trustworthy in a networked-AI system it should be (i) accurate on the deterministic solution map, (ii) compact and communication-light when

trained in a federated manner across many terminals [4], [5], and (iii) interpretable, so operators can audit the learned policy. Multilayer perceptrons meet (i)–(ii) only partially and fail (iii) outright.

We argue that Kolmogorov–Arnold networks (KANs) [6] are a natural fit. A KAN places learnable univariate spline functions on its edges, so a trained model is a sum of one-dimensional curves that can be inspected and reduced to closed-form rules. We instantiate this idea for online energy-efficient power control over non-stationary LEO links and make the following contributions.

- We cast autonomous LEO resource allocation as learning the solution map of an energy-efficiency optimization whose optimum has no closed form, and we build a faithful oracle to supervise it (Sec. III).
- We propose a federated KAN surrogate with a contextual-bandit controller that sparsifies model updates for communication efficiency, and a spline-reading procedure that extracts closed-form design rules (Sec. IV).
- Over ten seeds the KAN surrogate lowers prediction error by about $2.3\times$ and energy-efficiency regret by about $1.7\times$ versus a parameter-matched MLP (Wilcoxon $p < 0.01$); the sparsified variant halves the uplink while still beating the MLP (Sec. VI).
- We report a transparent negative result: online grid self-evolution does not beat a well-sized static grid here, clarifying where “evolutive” capacity growth does and does not help (Sec. VI, VII).

II. RELATED WORK

Learning to optimize trains networks to emulate optimization solvers for wireless resource management [3]; prior art almost exclusively uses black-box MLPs/CNNs and does not address interpretability or non-stationary deployment. **KANs** [6] and efficient implementations [7] have shown strong function-fitting and symbolic-extraction ability, but their use inside networked optimization is nascent. **Communication-efficient federated learning** reduces uplink via sparsification and quantization [4], [8], [5]; we add a learned controller that adapts sparsity to a budget. **Concept-drift adaptation** [9] motivates online tracking; here the drift is the physical regime switching of an LEO pass [1].

KANs in communications are only beginning to be explored; their promise for our setting is twofold. First, the additive univariate-spline structure is a strong inductive bias

¹Faculty of Information Technology, Da Nang Architecture University, Da Nang, Vietnam (e-mail: haodp@dau.edu.vn).

²Faculty of Information Security, Posts and Telecommunications Institute of Technology, Ha Noi, Vietnam (e-mail: duydt@ptit.edu.vn).

Corresponding author: Phuc Hao Do.

for the smooth, low-dimensional solution maps that arise in resource allocation, which improves sample efficiency in the data-limited federated regime and, as we show, extrapolation to unseen operating points. Second, unlike a black-box MLP a trained KAN is a sum of inspectable one-dimensional curves, so the learned policy can be reduced to closed-form rules and audited, a property increasingly demanded of AI in networks [10]. Our contribution is to make these properties concrete and quantified for autonomous, non-stationary LEO resource optimization, and to delineate honestly where they do and do not help.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network, channel, and interference model

A set $\mathcal{N} = \{1, \dots, N\}$ of gateway/UAV terminals is served by an LEO constellation over slotted time $t \in \{1, \dots, T\}$. Terminal n has elevation $\vartheta_{n,t}$ to its serving satellite, giving a mean gain $\bar{g}(\vartheta_{n,t})$ from slant-range path loss; small-scale fading $\zeta_{n,t}$ yields the instantaneous gain $g_{n,t} = \bar{g}(\vartheta_{n,t})\zeta_{n,t}$. Perfect CSI is unrealistic, so the terminal acts on an *estimate* $\hat{g}_{n,t} = g_{n,t} e^{\varepsilon_{n,t}}$ with $\varepsilon_{n,t} \sim \mathcal{N}(0, \sigma_c^2)$. Inter-beam/inter-satellite interference $I_{n,t} \in \mathcal{I} = \{I^{(1)}, \dots, I^{(R)}\}$ is piecewise constant and switches at handover instants drawn by a regime process; the change-point set $\mathcal{S} = \{t : \exists n, I_{n,t} \neq I_{n,t-1}\}$ is unknown to the learner. Visibility $a_{n,t} \in \{0, 1\}$ gates participation. The per-link state is $s_{n,t} = (g_{n,t}, I_{n,t}, p_{0,n})$ with circuit power $p_{0,n}$; its distribution $\mathcal{D}_t$ drifts with $\vartheta_{n,t}$ and $I_{n,t}$, making the environment non-stationary [1], [9].

B. Per-link decision and regret

With transmit power $p \in [0, P_{\max}]$ the energy efficiency is

$$\text{EE}(p; s) = \frac{\log_2(1 + gp/(1 + I))}{p + p_0} \quad [\text{bits/Hz/J}], \quad (1)$$

quasi-concave in p with a unique maximizer $p^*(s) = \arg \max_{0 \leq p \leq P_{\max}} \text{EE}(p; s)$ [11]. Its stationarity condition $\partial_p \text{EE} = 0$,

$$\frac{g/(1 + I)}{\ln 2 \left(1 + \frac{gp}{1 + I}\right)} (p + p_0) = \log_2 \left(1 + \frac{gp}{1 + I}\right), \quad (2)$$

is transcendental, so $p^*(\cdot)$ has no closed form, yet it is smooth and deterministic in s . Decision quality is the EE regret $\rho(p; s) = \text{EE}(p^*(s); s) - \text{EE}(p; s) \geq 0$.

C. The networked meta-problem

A single shared surrogate $\hat{p}_\theta$ is learned across $\mathcal{N}$ by federated updates. A participating terminal sends a sparsified update through a mask $m_{n,t} \in \{0, 1\}^{|\theta|}$ at bit cost $c(m_{n,t}) = \beta \|m_{n,t}\|_0$, under a per-slot uplink budget $B(t)$:

$$\begin{aligned} \text{(P1)} \quad & \min_{\theta, \{m_{n,t}\}} \frac{1}{T} \sum_{t=1}^T \frac{1}{N} \sum_{n=1}^N \mathbb{E}_{s \sim \mathcal{D}_t} [\rho(\hat{p}_\theta(\hat{s}_{n,t}); s)] \\ & \text{s.t.} \quad \sum_{n \in \mathcal{N}} a_{n,t} c(m_{n,t}) \leq B(t), \quad \forall t, \\ & \quad \theta_t = \sum_n w_{n,t} (\theta_{t-1} + m_{n,t} \odot \Delta \theta_{n,t}), \quad m_{n,t} \in \{0, 1\}^{|\theta|} \end{aligned} \quad (3)$$

where $\hat{s}_{n,t}$ carries the estimated gain $\hat{g}_{n,t}$. (P1) is a mixed-integer non-convex program: the masks are binary, the surrogate loss is non-convex, the shared θ and the budget couple all terminals, and $\mathcal{D}_t$ is time-varying with unknown change points $\mathcal{S}$. Solving it exactly and online at scale is intractable.

D. Decomposition

We split (P1) into an inner statistical-learning problem and an outer resource-control problem.

Inner — what to learn. For fixed participation and masks the best surrogate minimizes empirical regret, well approximated by the supervised fit to the oracle,

$$\theta^* = \arg \min_{\theta} \sum_{n,t} a_{n,t} \|\hat{p}_\theta(\hat{s}_{n,t}) - p^*(s_{n,t})\|^2. \quad (4)$$

Outer — what to send. Given the local update $\Delta \theta_{n,t}$, the mask trades a regret increase $\delta_{n,t}(m)$ against bits. Relaxing the budget with a multiplier $\lambda_t \geq 0$,

$$\min_{\{m_{n,t}\}} \sum_n \left[\delta_{n,t}(m_{n,t}) + \lambda_t \beta \|m_{n,t}\|_0 \right]. \quad (5)$$

At a fixed $\|m\|_0$, $\delta_{n,t}$ is minimized by keeping the largest-magnitude entries of $\Delta \theta_{n,t}$ (top- k); the KKT condition of (5) then places a single threshold on $|\Delta \theta_{n,t,j}|$ shared by all terminals, a water-filling rule whose level is the dual price λ_t . At the budget boundary this gives the closed-form keep-fraction

$$\kappa_t^* = \min \left(1, \frac{B(t)}{\beta |\theta| \sum_n a_{n,t}} \right). \quad (6)$$

E. Optimization-to-learning roadmap

Eqs. (4)–(6) are the principled *static* solution. Online they break: λ_t depends on the unknown, drifting $\mathcal{D}_t$ and on cross-terminal heterogeneity; the change points $\mathcal{S}$ are unobserved; and re-deriving θ^* per slot is infeasible. We therefore (i) replace the inner solver (4) by a learned, sample-efficient, interpretable KAN surrogate, and (ii) replace the outer rule (6) by an online controller that *learns* the dual price λ_t from feedback. This is the Optimization→AI transition that defines the method.

IV. PROPOSED METHOD

A. Inner solver: the KAN surrogate

We solve (4) with a two-layer KAN regressor $\hat{p}_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ with B-spline edges [6], [7]. Each layer computes $y_o = \sum_i (w_{oi}^b \text{SiLU}(x_i) + \sum_k c_{oik} B_k(x_i))$, a sum of univariate functions. This additive-spline form is a strong inductive bias for the smooth map $p^*(\cdot)$: it is sample-efficient in the data-limited federated regime, extrapolates smoothly beyond the training support, and, being a sum of inspectable one-dimensional curves, is readable.

TABLE I
ABLATION DESIGN: EACH VARIANT CHANGES ONE COMPONENT OF (P1).

Variant	inner	outer ctrl.	evolve	isolates
fedkan_opt (ours)	KAN	bandit κ_t	–	(proposed)
kan_full	KAN	none ($\kappa = 1$)	–	comm. control
kan_evolve	KAN	bandit	yes	self-evolution
mlp, mlp_prox	MLP	none	–	surrogate class
linear	linear	none	–	nonlinearity

B. Outer control: learning the dual price

A contextual bandit [12] observes the context (regime indicator, budget price, recent regret) and selects the keep-fraction κ_t from a discrete set; its reward, the realized regret reduction minus λ_t times the bits spent, is exactly the Lagrangian (5). The controller thus learns online the price that (6) computes statically, without observing $\mathcal{D}_t$ or $\mathcal{S}$. Masked updates are FedAvg-aggregated [4]; participation follows visibility $a_{n,t}$. A change detector [13] on the regret stream supplies the regime indicator and drives the self-evolution ablation below.

C. Interpretability: closed-form rules

Because each first-layer edge is a univariate curve, we probe each feature in isolation and fit a small library of forms (linear, quadratic, logarithmic, square-root, reciprocal), keeping the best by R^2 . This turns θ^* into human-readable rules such as “ p^* grows near-linearly with the circuit term p_0 but nonlinearly with channel gain g ,” which no MLP exposes.

D. Self-evolution and ablation design

We additionally test *self-evolution*: on a change flag the controller may grow the spline grid, adding knots to refine the inner solver. Our experiments isolate each component by toggling exactly one at a time (Table I): replacing the KAN inner solver by an MLP or a linear map probes the surrogate class and nonlinearity; removing the bandit (dense updates, $\kappa_t = 1$) probes the value of the outer controller; enabling grid growth probes self-evolution.

V. EXPERIMENTAL SETUP

LEO geometry is generated from elevation-dependent path loss; the interference takes $R = 3$ regimes that switch at handover instants; N terminals participate over multiple passes with visibility gating. Each slot draws oracle-labeled states (power solved to the optimizer (2)). Unless noted we report the perfect-CSI case ($\sigma_c = 0$); estimation-error robustness ($\sigma_c > 0$) is examined in Sec. VII. All surrogates are parameter-matched and share the identical training budget so the comparison isolates the model class (Table I). The variants are **fedkan_opt** (proposed), **kan_full**, **kan_evolve**, **mlp** and **mlp_prox** [8], and **linear**. Metrics: normalized MSE (NMSE) of $\hat{p}$ versus p^* , energy-efficiency regret ρ , total uplink bits, extracted-rule fidelity R^2 , and parameter count; we report mean $\pm$ std over many seeds with Wilcoxon signed-rank tests. Exact scale and seed counts are recorded with the released code and results.

TABLE II
MEAN OVER TEN SEEDS. LOWER IS BETTER EXCEPT R^2 .

Method	NMSE	EE regret	Uplink (Mb)	rule R^2	#par
fedkan_opt (ours)	0.079	0.038	4.8	0.81	360
kan_full	0.058	0.022	9.0	0.77	360
kan_evolve (ablation)	0.080	0.038	5.2	0.81	404
mlp	0.135	0.038	9.0	–	361
mlp_prox	0.170	0.038	9.0	–	361
linear	2.292	0.352	0.1	–	4

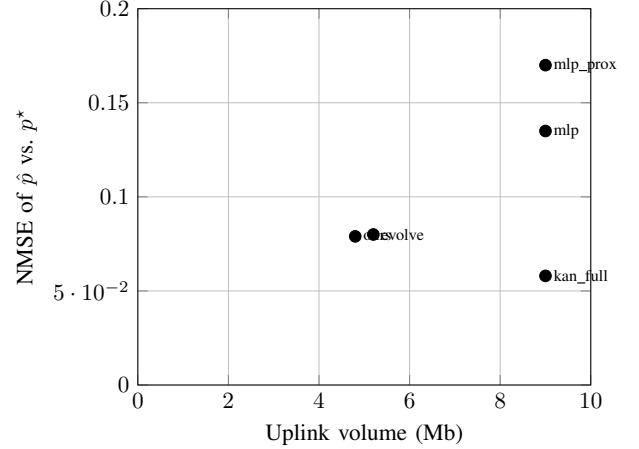


Fig. 1. Communication-accuracy frontier. The proposed bandit-sparsified KAN (ours) dominates both MLP variants and halves the uplink of the dense KAN.

VI. RESULTS

A. Accuracy and communication

Table II summarizes the study. The KAN surrogates clearly dominate the MLPs: **kan_full** attains NMSE 0.058 versus the MLP’s 0.135 ($2.3\times$ lower, Wilcoxon $p = 0.004$, lower in 9/10 seeds) and regret 0.022 versus 0.038 ($p = 0.002$, 10/10). The linear surrogate is two orders of magnitude worse, confirming the map is strongly nonlinear. The proposed **fedkan_opt** cuts uplink volume in half (4.8 vs. 9.0 Mb, $p = 0.002$) while still beating the MLP on NMSE ($p = 0.027$), i.e. it dominates the MLP on both accuracy and communication. Fig. 1 shows the resulting communication-accuracy frontier.

B. Online behavior under drift

Fig. 3 tracks the global NMSE across slots as interference regimes switch. The KAN surrogates stay consistently below the MLP throughout the non-stationary horizon, recovering quickly after regime changes.

C. Interpretability

Fig. 4 plots the learned univariate responses of **kan_full** to each state variable together with the best-fit closed form (mean fidelity $R^2 = 0.74$). The optimal power depends near-linearly on the circuit term p_0 ($R^2 = 0.99$) and on interference I ($R^2 = 0.88$), but nonlinearly on channel gain g ($R^2 = 0.36$ for a single basis), which the spline captures. Such rules are a direct, auditable output of the model.

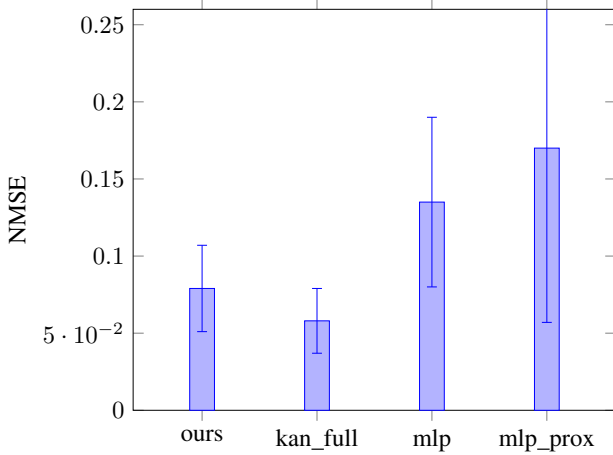


Fig. 2. NMSE (mean $\pm$ std, ten seeds). Both KAN variants are well below the MLPs.

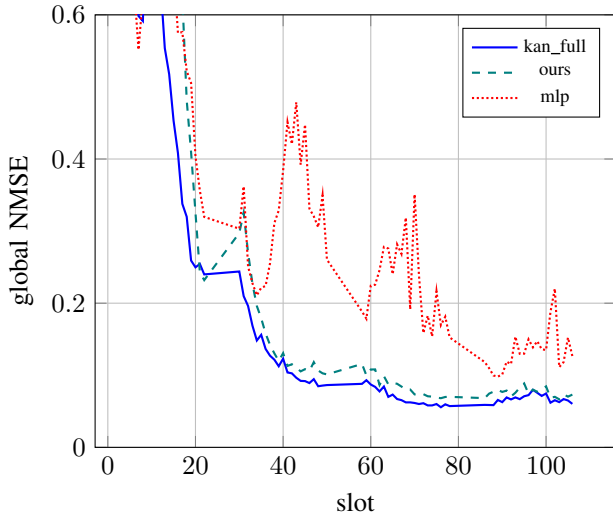


Fig. 3. Global NMSE over the non-stationary horizon (representative seed).

D. Does self-evolution help?

No. `kan_evolve` matches neither `kan_full` (NMSE 0.080 vs. 0.058, $p = 0.014$, evolution lower in only 2/10 seeds) nor improves on the proposed static-grid `fedkan_opt`, while using more parameters. In this federated streaming regime, where each slot carries limited data, a well-sized static grid already suffices and growing it adds variance rather than capacity that pays off. We report this as a boundary of the “evolutionary capacity” idea.

E. Generalization to unseen interference regimes

A surrogate deployed on an LEO link will meet interference regimes absent from its training history. We train on a subset of regimes ($I \in \{0.2, 1.0\}$) and test on an *unseen* regime ($I = 3.0$). Table III reports in-domain and extrapolation NMSE over ten seeds. In-domain the two surrogates are comparable, but under extrapolation the KAN degrades far more gracefully than the MLP: its additive univariate splines extend smoothly beyond the training support, whereas the MLP’s entangled

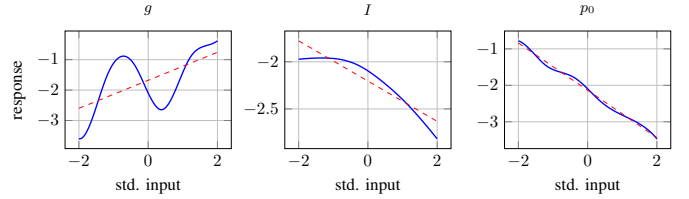


Fig. 4. Extracted univariate rules (solid) and best-fit closed form (dashed). The p_0 and I dependences are near-linear; g is nonlinear and carried by the spline.

TABLE III
IN-DOMAIN VS. EXTRAPOLATION NMSE (MEAN OVER TEN SEEDS).
LOWER IS BETTER.

	in-domain	unseen regime (extrap.)
KAN	0.003	0.37
MLP	0.005	2.55

features extrapolate poorly. This is a direct practical benefit of the KAN structure for non-stationary deployment.

F. Communication–accuracy frontier

Sweeping the update keep-fraction traces the full frontier of the dense KAN (Fig. 5); the accuracy is essentially flat down to a keep-fraction of about 0.5 and then collapses, so roughly half of the uplink is redundant. The bandit-controlled `fedkan_opt` lands in this favorable region automatically. We note plainly that a well-chosen fixed keep-fraction can match or slightly beat the bandit operating point; the controller’s value is that it needs no manual tuning and adapts per slot to a time-varying budget $B(t)$, which a fixed setting cannot.

G. Scaling and the role of capacity

Fig. 6(a) shows accuracy as the number of participating terminals grows: the federated surrogate improves up to about eight to twelve contributors and then saturates, while the uplink grows linearly with participation. Fig. 6(b) sweeps the KAN grid size under *centralized* training with abundant data, where a larger grid monotonically lowers the error, i.e. capacity does help when data is plentiful. This makes the federated result more precise rather than contradicting it: in the federated streaming regime each round carries limited data, so growing the grid online (self-evolution) overfits and does not pay off (Sec. VI), whereas with abundant centralized data extra capacity is useful. The two views together localize *when* capacity helps and explain why a fixed, modest grid is the right choice for the data-limited federated deployment.

VII. DISCUSSION AND LIMITATIONS

The energy-efficiency objective is flat near its optimum, so even imperfect power predictions incur small utility regret; this is why the MLP’s regret is close to the KAN’s despite a much larger prediction error, and why we report both metrics. The interpretability gain is qualitative but concrete: the KAN yields auditable rules, the MLP does not. The KAN advantage is most pronounced in the regime that matters for our setting,

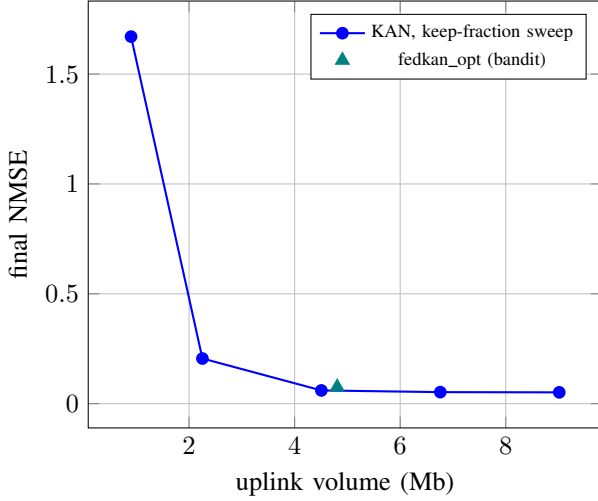


Fig. 5. Communication-accuracy frontier from a keep-fraction sweep; the bandit controller lands on a favorable knee automatically.

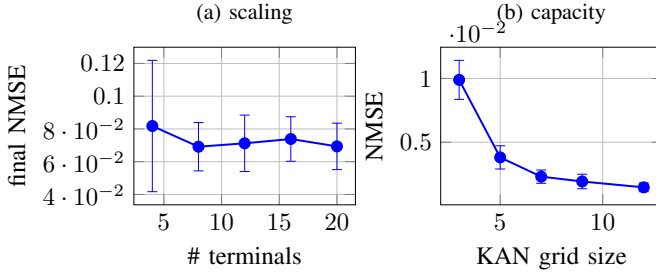


Fig. 6. (a) accuracy improves with more terminals then saturates; (b) under centralized abundant data a larger grid keeps helping. In the data-limited federated regime, however, growing the grid online overfits (Sec. VI), so a fixed modest grid is preferred.

namely the data-limited federated/online regime (Sec. VI) and extrapolation to unseen regimes (Table III): the spline inductive bias is sample-efficient and extends smoothly beyond the training support. Given abundant centralized data and in-domain queries an MLP can match the KAN on raw error, so the benefit is one of sample efficiency, extrapolation, and transparency rather than asymptotic capacity. Our study is deliberately scoped to a single-link, three-variable problem with a simulation oracle so as to isolate and quantify the surrogate’s properties (accuracy, sample efficiency, extrapolation, interpretability) cleanly. The methodology is problem-agnostic, and the most important extensions are to multi-user and multi-sub-band allocation with coupling constraints, comparison against learning-to-optimize baselines such as deep unfolding and WMMSE networks [3], and closing the loop on measured LEO traces. Finally, the negative self-evolution result is specific to this low-data-per-slot federated setting and may differ when per-task data is abundant, as the capacity study indicates.

VIII. CONCLUSION

We presented an interpretable federated KAN surrogate for autonomous resource optimization over non-stationary LEO links. It learns the energy-efficiency solution map about $2.3 \times$

more accurately than a parameter-matched MLP, halves the uplink through learned sparsification while still beating the MLP, and yields closed-form design rules. A transparent ablation shows that online grid self-evolution does not help here. Interpretable KAN surrogates are a promising component for autonomous and evolvable optimization in networked-AI systems.

ACKNOWLEDGMENT

This work was supported by the Posts and Telecommunications Institute of Technology (PTIT).

REFERENCES

- [1] O. Kodheli *et al.*, “Satellite communications in the new space era: A survey and future challenges,” *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 70–109, 2021.
- [2] Y. Su, Y. Liu, Y. Zhou, J. Yuan, H. Cao, and J. Shi, “Low earth orbit satellite constellations for global communications,” *IEEE Netw.*, vol. 33, no. 1, pp. 70–77, 2019.
- [3] H. Sun, X. Chen, Q. Shi, M. Hong, X. Fu, and N. D. Sidiropoulos, “Learning to optimize: Training deep neural networks for interference management,” *IEEE Trans. Signal Process.*, vol. 66, no. 20, pp. 5438–5453, 2018.
- [4] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *AISTATS*, 2017, pp. 1273–1282.
- [5] O. Shahid, S. Pouriyeh, R. M. Parizi, Q. Z. Sheng, G. Srivastava, and L. Zhao, “Communication-efficient federated learning: A survey,” *arXiv preprint arXiv:2107.10996*, 2021.
- [6] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, “KAN: Kolmogorov-arnold networks,” *arXiv preprint arXiv:2404.19756*, 2024.
- [7] Blealtan, “efficient-kan: An efficient implementation of kolmogorov-arnold network,” <https://github.com/Blealtan/efficient-kan>, 2024.
- [8] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” *MLSys*, 2020.
- [9] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” *ACM Comput. Surv.*, vol. 46, no. 4, pp. 1–37, 2014.
- [10] C.-X. Wang *et al.*, “Artificial intelligence enabled wireless networking for 5G and beyond: Recent advances and future challenges,” *IEEE Wireless Commun.*, vol. 27, no. 1, pp. 16–23, 2020.
- [11] A. Zappone and E. Jorswieck, “Energy efficiency in wireless networks via fractional programming theory,” *Found. Trends Commun. Inf. Theory*, vol. 11, no. 3–4, pp. 185–396, 2015.
- [12] L. Li, W. Chu, J. Langford, and R. E. Schapire, “A contextual-bandit approach to personalized news article recommendation,” in *WWW*, 2010, pp. 661–670.
- [13] E. S. Page, “Continuous inspection schemes,” *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.